GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 10



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Recap of Previous Lecture



Topic Counting Theory

Topics to be Covered



Topic-1 Time and Work

[MCQ]

#Q. Find the least perfect cube divisible by 2, 3, 4 and 6.

Assignment

A 216

B 1728

C 512

D 360

[MCQ]

#Q. How many natural numbers up to 100 are divisible by both 2 and 3?

6

Assignment

A 13

B 14

C 17

D 16

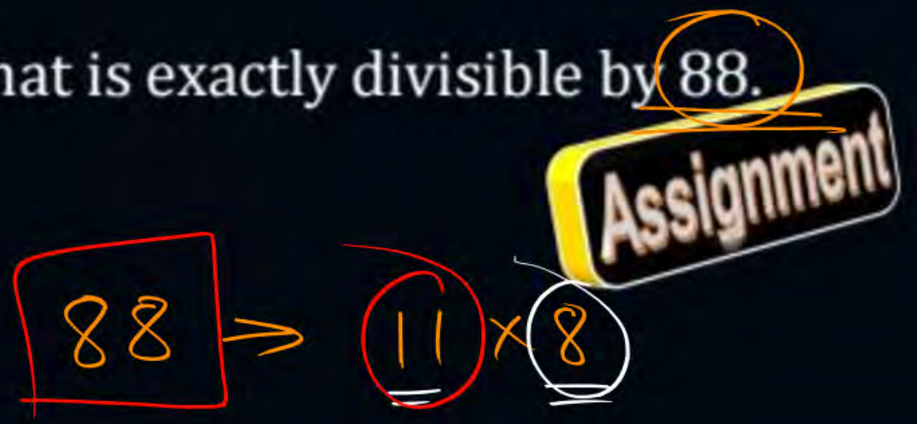
$6 \times 1 \Rightarrow 6$

2 12

$6 \times 16 \Rightarrow 96$

[MCQ]

#Q. Find the largest four-digit number that is exactly divisible by 88.



A 9844 ✗

B 9768 ✓

C 8894 ✗

D 9944 ✓

[MCQ]

#Q. In how many ways can you arrange 5 Novels on a shelf?

Assignment

A 720

B 120

C 360

D 300

$5 \times 4 \times 3 \times 2 \times 1 = 120$

N_1 N_2 N_3 N_4 N_5

N_1 N_2 $\vdots$ N_5

$(5-2)$ $(5-3)$ $(5-4)$ $(5-1)$

[MCQ]

#Q. Four persons A, B, C and D are to be seated in a row. C should not be seated at the second position from the left end of the row. The number of distinct seating arrangements possible is?

Assignment

$3 \times 3 \times 2 \times 1 = 18$

(4-1) (4-2) (4-3)

A
B
C
D

A
B
C
D

- A. 6
- B. 9
- C. 18
- D. 24

[MCQ]

#Q. Five different book (P, Q, R, S, T) are to be arranged on a shelf. The books R and S are to be arranged first and second, respectively from the right side of the shelf. The number of different orders in which P, Q and T may be arranged is?

Assignment

A 12

B 120

C 6

D 2

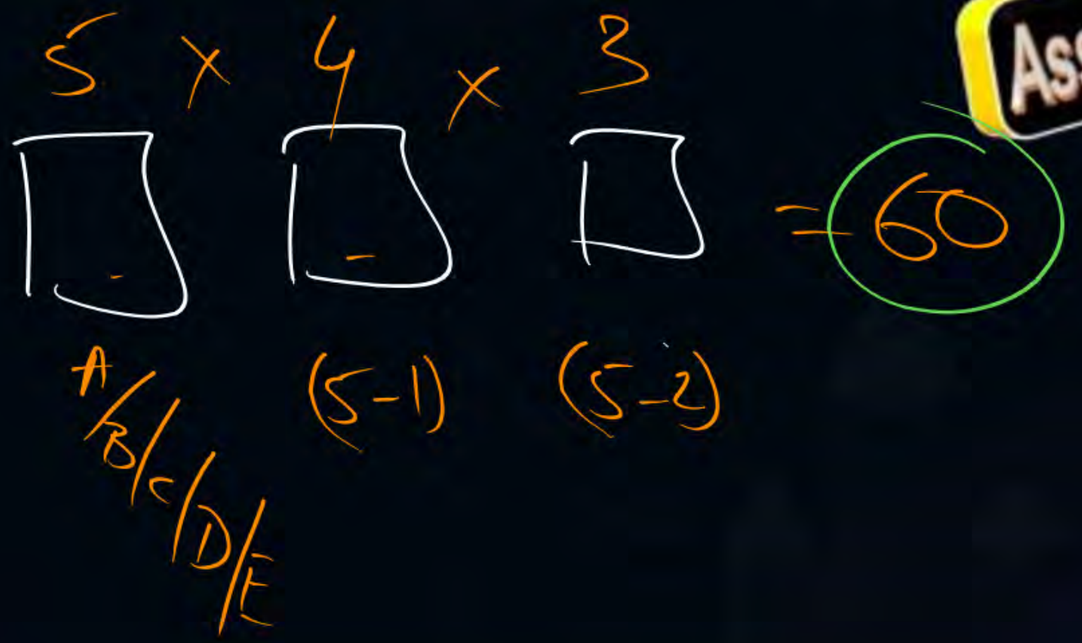
$3 \times 2 \times 1 \times \underset{1}{S} \times \underset{1}{R} = 6$

P/Q/T P/Q/T

3-1

[MCQ]

#Q. A license plate begins with 3 letters. If the possible letters are A, B, C, D and E, how many different ways these letters can be written if no letter is used more than once?



A 720

B 120

C 60

D 30

[MCQ]

#Q. In how many different ways can the letters of the word 'PLEADING' be arranged in such a way that the vowels always come together?

Assignment

and also the consonants

A 720

B 4320

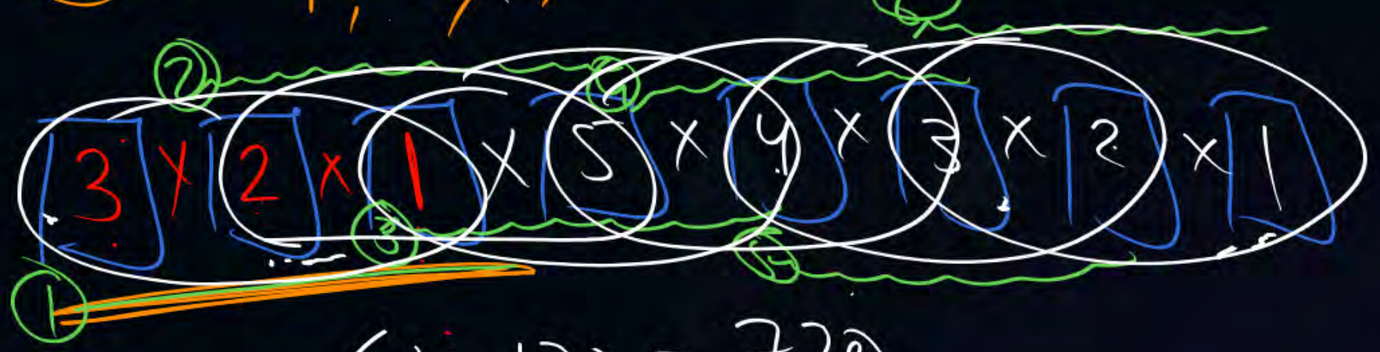
C 5040

D 3600

720
× 6
4320

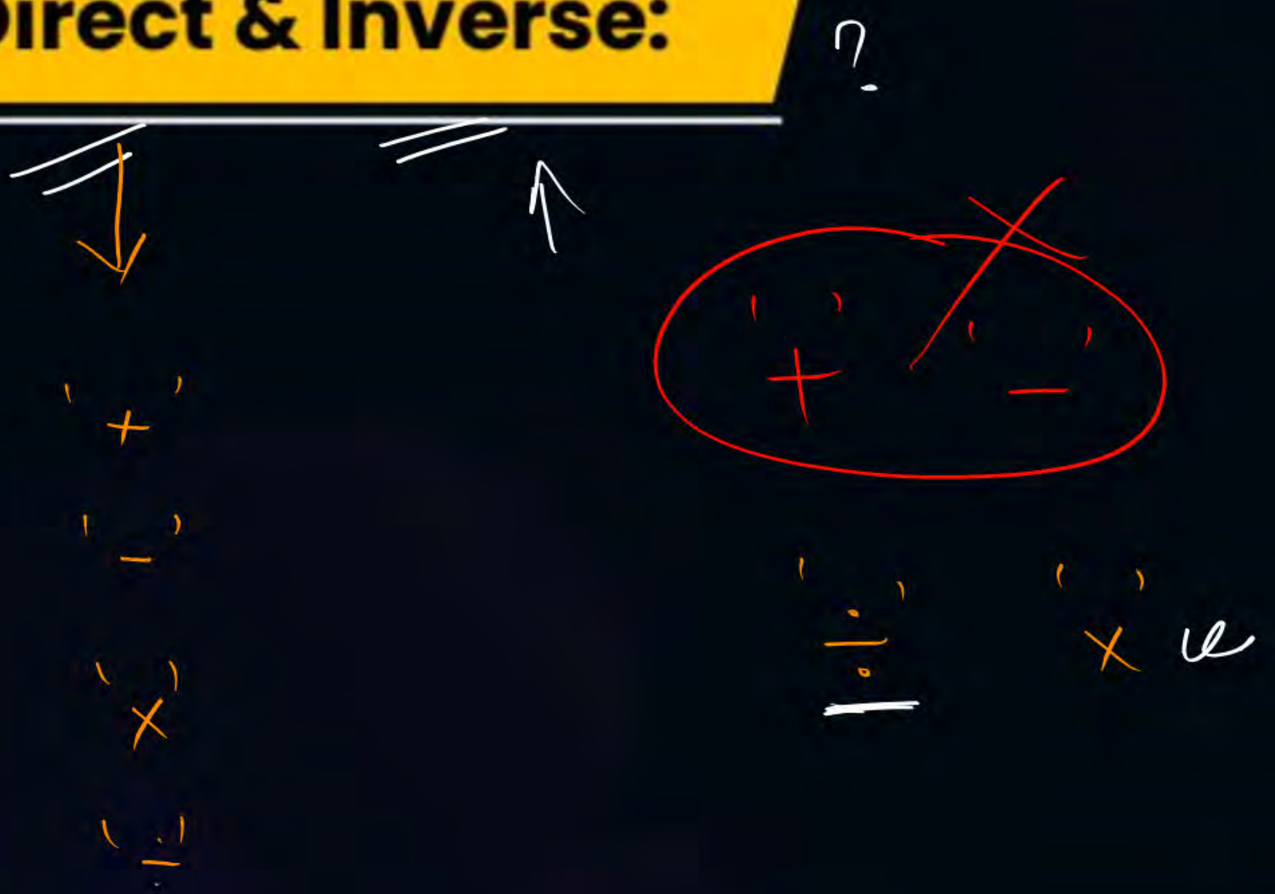
V ⇒ E, A, I

C ⇒ P, L, D, N, G

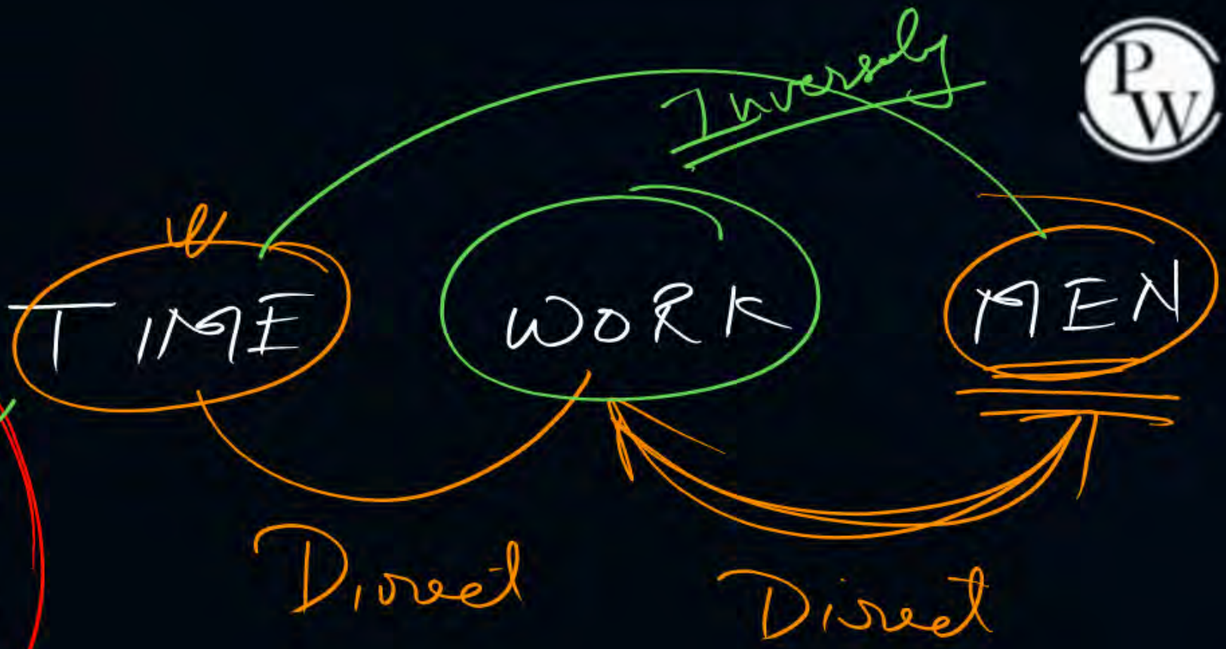


6 × 120 = 720

Direct & Inverse:



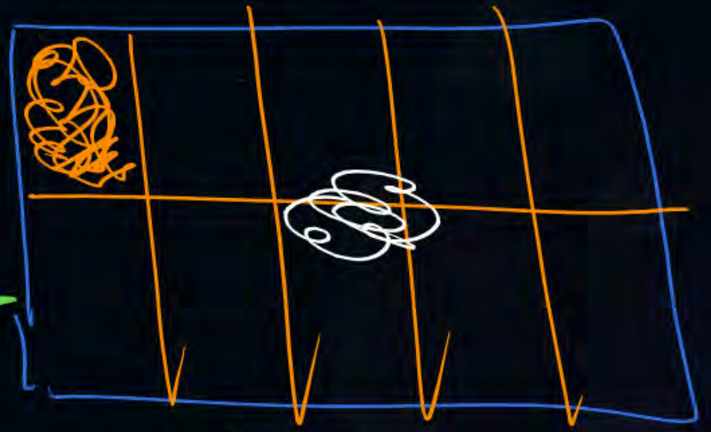
Time & Work



A = 10 days $\Rightarrow \frac{1}{10}$

B = 20 days $\Rightarrow \frac{1}{20}$

A & B $\Rightarrow \frac{1}{10} + \frac{1}{20} = \frac{3}{20}$



$\frac{20}{3} \Rightarrow 6.\bar{6}$ days OR $6\frac{2}{3}$ days

Let's Try:

Q. M ⇒ 60 days

M & N ⇒ 40 days

N ⇒ ? 120 days

Q. P ⇒ 20 days ; Q ⇒ 30 days

P & Q ⇒ 12 days

Q. A ⇒ 25 days ; B ⇒ 40 days

A & B ⇒ $15\frac{5}{13}$ days

OR

15.38 days

[MCQ]

$A = \frac{1}{10}$ | $B = \frac{1}{20}$

#Q. A can do a work in 10 days whereas B in 20 days.They started working on alternate days, i.e. 1st day 'A', 2nd day 'B', 3rd day 'A'.....so on. Then in how many days the total work would be completed ?

$\frac{?}{20} = \frac{1}{10}$?

$\frac{1}{10} + \frac{1}{20} = \frac{3}{20}$ ×6 ⇒ 2 days

13 days ∴

⇒ $\frac{18}{20}$ ⇒ 12 days



[MCQ]

$A = \frac{1}{10}$ | $B = \frac{1}{20}$

#Q. A can do a work in 10 days whereas B in 20 days.They started working on alternate days, i.e. 1st day 'B', 2nd day 'A', 3rd day 'B'.....so on. Then in how many days the total work would be completed ?

$\frac{2}{20} = \frac{1}{10}$?

13th day

$\frac{1}{10} - \frac{1}{20} = \frac{1}{20}$

1 work A = 10

$\frac{1}{20}$ " $A = \frac{1}{20_2} \times 10 = \frac{1}{2}$

1	2	3	4	5	6	7	8	9	10	11	12	13
B	A	B	A	B	A	B	A	B	A	B	A	B

$\frac{1}{20} + \frac{1}{10} = \frac{3}{20} \Rightarrow 2$ days

$\frac{18}{20} \Rightarrow 12$ days

$13\frac{1}{2}$ days

[MCQ]

#Q. A can do a work in 25 days whereas B in 40 days. They started working on alternate days, i.e. 1st day 'A', 2nd day 'B', 3rd day 'A'.....so on. Then in how many days the work would be completed?

A. $15\ ^{5}/_{8}$ days

B. $30\ ^{5}/_{8}$ days

C. 31

D. 16

[MCQ]

20 days

$A = \frac{1}{20}$ | $B = \frac{1}{30}$ | $C = \frac{1}{60}$

#Q. A can do a work in 20 days whereas B in 30 days and C in 60 days. They started the work together. But A left after two days and B left three days before the work got completed. Then the total work was done in how many days?

A&B&C

$\Rightarrow \frac{1}{20} + \frac{1}{30} + \frac{1}{60}$

$= \frac{6}{60} = \frac{1}{10}$

B&C $\rightarrow \frac{1}{30} + \frac{1}{60} = \frac{3}{60} = \frac{1}{20}$

$\frac{2}{10}$ A&B&C	$\frac{3}{4}$ B&C	$\frac{3}{60}$ C $= \frac{1}{20}$
2 days	15 days	3 days

$\frac{2}{10} + \frac{1}{20} = \frac{3}{20} = \frac{1}{4}$

1 work B&C = 20 days

$\frac{3}{4}$ " " $= 20 \times \frac{3}{4}$

$= 15$

[MCQ]

'x' days

#Q. A can do a work in 20 days whereas B in 30 days and C in 60 days. They started the work together. But A left after two days and B left three days before the work got completed. Then the total work was done in how many days?

$\frac{x}{60} + \frac{x-3}{30} + \frac{2}{20} = 1$

$x + 2x - 6 + 6 = 60$

$3x = 60$ 20

[MCQ]

27 - 5 = 22 u

$A = \frac{1}{10}$ | $B = \frac{1}{20}$ u | $C = \frac{1}{60}$ u

#Q. A can do a work in 10 days whereas B in 20 days and C in 60 days. They started the work together. But A left at the end of 3rd day and B left at the end of 5th day. Then the remaining work was done by C in how many days?

3 days	2 days	? ? days
A&B&C	B&C	C

x' days

$\frac{x}{60} + \frac{5}{20} + \frac{3}{10} = 1$

$\Rightarrow x + 15 + 18 = 60$

$x = 60 - 33 = 27$

A 27

B 35

C 31

D 22

[MCQ]

#Q. If 9 MONKEYS EAT 9 BANANAS IN 9 MINUTES, THEN HOW MANY MONKEYS WILL EAT 45 BANANAS IN 45 MINUTES?

Assignment



2 mins Summary

Topic ~~Counting Theory~~, Time and Work

THANK - YOU